Literature status: the c_0 spreading-model question for K^s

Run `fa_banach_001`; source arXiv:1711.05149

Source question

Hajek, Kania, and Russo's paper *Symmetrically separated sequences in the unit sphere of a Banach space* asks the following question about spreading models and the symmetric Kottman constant.

> Concerning spreading models, we have proved above that $K^s(X) = 2$ whenever X admits a spreading model isomorphic to ℓ_1. We do not know whether an analogous result for c_0 holds true too.
>
> **Problem 5.11.** Suppose that a Banach space X admits a spreading model isomorphic to c_0. Does it follow that $K^s(X) = 2$?

Figure 1: Problem 5.11 in arXiv:1711.05149, page 17.

The question is whether

$$X \text{ admits a spreading model isomorphic to } c_0 \quad \Longrightarrow \quad K^s(X) = 2.$$

Supporting literature status

Russo's later note *A note on symmetric separation in Banach spaces* explicitly identifies this question and says that its Theorem B solves [1, Problem 5.11] negatively.

> In the second main result of this note we show that, indeed, the problem with vectors having 'long support' can not be circumvented. The result, whose formal statement in given in Theorem B below, solves in the negative [14, Problem 5.11] in a very strong way.

Figure 2: arXiv:1904.12598, page 2: Theorem B is stated to solve [1, Problem 5.11] negatively.

> **Theorem B.** *For every $\varepsilon > 0$ there exists a Banach space X every whose spreading model is isomorphic to c_0 and such that $K(X) \leqslant 1 + \varepsilon$.*

Figure 3: arXiv:1904.12598, page 3: Theorem B.

Russo also records the concrete computation behind Theorem B for the original Tsirelson space T_θ^*.

Proof intuition

The source problem is tempting because c_0 itself has symmetrically 2-separated unit vectors built from long initial-support patterns. A spreading model, however, preserves fixed finite patterns asymptotically; it does not automatically transfer constructions whose support length grows with

We shall now compute the (symmetric) Kottman's constant of T_θ^*.

Theorem 4.2. *Let $\theta \in (0,1)$ and T_θ^* be the original Tsirelson's space defined above. Then*

$$K(T_\theta^*) = K^s(T_\theta^*) = \begin{cases} 2 & \textit{if } \theta \in (0,1/2] \\ 1/\theta & \textit{if } \theta \in (1/2,1). \end{cases}$$

At the risk of stating the obvious, let us note that Theorem B is a direct consequence of Theorem 4.2. We could derive the proof of the result under consideration from the already mentioned result by Maluta and Papini concerning reflexive Banach spaces with the non-strict Opial property; however, we prefer to offer a direct argument, based on the suppression 1-unconditionality. As the reader will see, the proof is based on a sliding hump argument, somewhat similar to Kottman's proof ([18]) that $K(\ell_p) = 2^{1/p}$, for $1 \leqslant p < \infty$.

Proof. Let us start with a lower bound for $K^s(T_\theta^*)$, which is simply obtained by checking

Figure 4: arXiv:1904.12598, page 7: Theorem 4.2 and the note that Theorem B is a direct consequence.

the index. Russo makes this obstruction concrete with the original Tsirelson spaces. Their spreading models are all isomorphic to c_0, but the asymptotic c_0 structure still imposes an upper bound on ordinary separated sequences.

The literature answer

Theorem 1. *Problem 5.11 of [1] has a negative answer. More precisely, for every $\varepsilon \in (0,1)$ there is a Banach space X such that every spreading model of X is isomorphic to c_0, but $K^s(X) < 2$.*

Proof. Russo's Theorem B [2] states that for every $\varepsilon > 0$ there exists a Banach space X, all of whose spreading models are isomorphic to c_0, such that $K(X) \le 1 + \varepsilon$. Since symmetric σ-separation implies ordinary σ-separation, one always has

$$K^s(X) \le K(X).$$

Choosing $\varepsilon \in (0,1)$ gives

$$K^s(X) \le K(X) \le 1 + \varepsilon < 2,$$

while every spreading model of X is isomorphic to c_0. This is exactly the negation of the conclusion asked in Problem 5.11.

For a concrete family, Russo's Theorem 4.2 computes the constants of the original Tsirelson space T_θ^*:

$$K(T_\theta^*) = K^s(T_\theta^*) = \begin{cases} 2, & 0 < \theta \le 1/2, \\ 1/\theta, & 1/2 < \theta < 1. \end{cases}$$

The same section recalls that every spreading model of T_θ^* is isomorphic to c_0. Hence any $\theta \in (1/2,1)$ gives $K^s(T_\theta^*) = 1/\theta < 2$, and θ close to 1 gives Russo's arbitrarily small upper bound $K(T_\theta^*) \le 1 + \varepsilon$. □

Verification notes and limitations

This packet is a literature-status identification, not a new proof. The supporting paper explicitly cites the source problem as [1, Problem 5.11] and states that the answer is negative. The inequality $K^s(X) \le K(X)$ is immediate from the definitions: symmetric separation includes separation for differences.

The result answers only the c_0 spreading-model question, Problem 5.11. It does not settle the broader Problem 5.12 in [1], which asks whether $K^s(X) \ge K^s(Z)$ for arbitrary spreading models Z of X.

The run lightweight indexes were searched for arXiv:1711.05149, the title, and the key phrases "spreading model", "c_0", and "Kottman" before promotion. No existing exact packet or attempt for this question was found. The exact supporting source was found by primary-source arXiv/web search for Kottman constants and c_0 spreading models.

Human review recommendation

Treat arXiv:1711.05149 Problem 5.11 as already answered negatively by arXiv:1904.12598. This packet should prevent duplicate agent work on the same problem, but it should not be counted as an original proof from this run.

References

[1] P. Hajek, T. Kania, and T. Russo, *Symmetrically separated sequences in the unit sphere of a Banach space*, J. Funct. Anal. 275 (2018), 3148–3168; arXiv:1711.05149.

[2] T. Russo, *A note on symmetric separation in Banach spaces*, arXiv:1904.12598.